Charles “Ben” Pratt

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EDUCATION

University of Florida - Gainesville, FL — Expected May 2027

B.S. Computer Science, Minor in Digital Arts and Sciences | GPA: 3.85

SKILLS

Reinforcement Learning	PPO, Actor-Critic, Reward Design, Domain Randomization, Sim-to-Real Transfer
Simulation & ML	Isaac Lab, Isaac Gym, MuJoCo, PyTorch, Diffusion Policy, Imitation Learning, GR00T, SmolVLA
Robotics & Control	ROS 2, Cartesian Impedance Control, Inverse Kinematics, Contact Dynamics, State Estimation, SLAM, Motion Planning, Humanoid & Legged Locomotion
Perception	RealSense RGB-D, LiDAR, Elevation Mapping, OpenCV, Sensor Calibration
Software	Python, C++, Java, C, C#, Rust, TypeScript, Linux, Git, CI/CD, Docker, AWS

WORK EXPERIENCE

RL Locomotion Intern Persona AI - Pensacola, FL — Aug 2026 - Present

- Adapted ray-based foot-placement rewards in an existing humanoid walking environment; course completion improved from 80% to 95% over 1,000 simulation episodes per policy, with slightly slower walking.
- Trained a policy that was tested on hardware, where the humanoid ascended and descended a full flight of stairs.
- Compared two perception and training approaches: a depth-image CNN distilled from a privileged height-map teacher, and a height-map MLP trained with actor-critic learning and a noise curriculum.
- In 100 simulation trials per policy, depth and height-map approaches reached the destination at an average speed of at least 0.54 m/s in 95% and 90% of trials, respectively; neither policy fell.
- Traced lost grated-stair geometry to depth-image resizing by comparing recorded sensor footage with policy inputs.
- Evaluated an additional LiDAR/depth policy in simulation: it improved grated-stair ascent and descent but performed worse on ordinary stairs.

Robotics Research Intern IHMC - Pensacola, FL — Summers 2025 & 2026

- 2026: Built Isaac Lab lever-turning tasks and a GR00T training pipeline using approximately 20 real and 200 simulated demonstrations; wrote a Java log converter and integrated Isaac Teleop's simulation-data converter.
- Adapted NVIDIA inverse-kinematics code and corrected a quaternion-convention mismatch across simulators; lever turning succeeded in 95% of 1,000 simulation trials and in one hardware trial.
- 2025: Collected demonstrations, trained LeRobot diffusion policies, and deployed them on the Unitree H1-2 through a Python-Java/ROS 2 integration.

PROJECTS & LEADERSHIP

Design Team Captain Gator TPED — Aug 2024 - May 2026

- Wrote Mosquitto MQTT communication and IR sensor sequencing for an inverted dark ride prototype; supported another student's track-actuation work.
- Helped lead approximately 50 students; proposed subsystem work sessions and helped introduce full-group design reviews every fourth meeting to resolve integration issues.

SmolVLA Block Placement UF Autonomous Robots — 2025

- Trained SmolVLA on 200 SO-101 leader-follower demonstrations to pick a requested colored block and place it at the center of a coaster.
- Built ROS 2 nodes to pass LLM task instructions and camera images to the VLA policy and execute its predicted arm actions.

Animatronic Gator Head Gator TPED

- Built a PySide6 interface for manual control and CSV animation export/playback, with serial commands driving jaw, eye, and eyebrow servos through Arduino and a PCA9685 controller.
- Implemented FFT-based audio analysis to map audio amplitude to jaw motion, making the gator appear to talk.